Ryan Quinn

rquinn1@mit.edu | rquinnmit.github.io | github.com/rquinnmit | linkedin.com/in/ryan-quinn-521139299

Education

Massachusetts Institute of Technology (MIT) Expected May 2027
Bachelor of Science in Artificial Intelligence & Decision Making

- **Coursework:** Reinforcement Learning, Computer Vision, Computer Graphics, Robotic Manipulation, Machine Learning, Computational Cognitive Science, Probability and Inference, Ethics of Computing
- **Extracurricular:** SERC Scholars Program; MIT Entrepreneurship Club; Chi Phi Fraternity (House Manager); Varsity Lacrosse (2 years)

Experience

MIT Computer Science & Artificial Intelligence Lab (CSAIL) January 2025 – Present
Machine Learning Undergraduate Researcher

- Leading research on multimodal 4D reconstruction for marine robots, an open problem that involves fusing video, DVL, IMU, and depth sensor inputs into a GTSAM pose graph to build point maps in hazy, hostile underwater scenes.
- Benchmarked MapAnything, VGGT4D, Any4D, and Depth Anything 3 on rendered and real underwater footage to select a backbone for further fine-tuning.
- Extending MapAnything to 4D with a motion estimator that uses optical flow, gyroscope, and DVL readings to separate moving objects from camera motion.
- Built a differentiable physics simulator that supplies physical constraints to 3D generative diffusion models.
- Developed flowSim (C/Python), a network simulator with multi-class traffic that generated millions of labeled flows to train m3, an ML model predicting flow completion times.

Coinbase May 2026 – August 2026
Software Engineer Intern

- Built a central registry of rules for AI coding agents, giving engineers one place to define what agents may do in Coinbase infrastructure.
- Shipped a Model Context Protocol (MCP) service that lets engineers and contractors create, share, and update prompts, skills, and agent specifications directly from their IDEs and coding agents.
- Built a service that flags malformed and low-quality records in the agent pipeline and alerts the owning engineers.

Amazon May 2025 – August 2025
Software Development Engineer Intern

- Launched two new recommendation strategies for the Amazon Shopping desktop homepage (~1000 impressions/sec), driving $1M+ in incremental annual revenue.
- Cut feature-store lookup latency by 86% (35 to 5 ms), speeding up homepage recommendations and saving ~$50,000/month in cloud spend.

Projects

VisionBeam (Autonomous Lighting System) June 2025 – October 2025

- Built a real-time system that tracks people with YOLOv8 and ByteTrack and steers motorized stage lights (DMX) at 40 FPS, holding identities through occlusions in crowded scenes.
- Designed an activity-scoring heuristic (bounding boxes + frame differencing) that builds spatial heatmaps of peak motion to select the highest-activity target automatically.

Skills

Languages: Python, C++, C, Julia, Java, Go, TypeScript, JavaScript, C#, SQL
ML, Vision & Graphics: PyTorch, JAX, TensorFlow, Gen.jl, CUDA, NumPy, SciPy, OpenCV, gsplat, GTSAM, Blender, Unity
Backend & Web: Flask, Node.js, React, Next.js, MySQL, DynamoDB, OpenAPI/Swagger, Dapper, HTML, CSS
Infrastructure & Tooling: AWS, Docker, Kubernetes, Linux, Model Context Protocol (MCP), Claude Code, Cursor